Kainat Riaz

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AI researcher working on LLM agents: test-time reinforcement learning, epistemic uncertainty, and human-in-the-loop decision-making for agents.

Education

National University of Sciences and Technology (NUST) 2020 – 2022
MS Electrical Engineering (AI & Autonomous Systems)

- CGPA 3.90/4.00, 3rd position on merit. Thesis: Targeted Adversarial Attack on Classic Vision Pipelines (published in CVIU, 2024).

Air University 2016 – 2020
BS Electrical Engineering (Electronics)

- CGPA 3.42/4.00. Merit scholar with a merit-based tuition fee waiver.

Publications

Under review

- **Riaz, K.***, Mohsin, M.A.*, Bilal, A., Umer, M., Mohsin, A., Riaz, A., Subhan, A., Cioffi, J.M. "Epistemic Uncertainty for Test-Time Discovery." arXiv:2605.11328, under review at ICLR 2027. (*equal contribution; collaboration with Stanford University)

Published

- **Riaz, K.**, Anjum, M.L., Hussain, W., Manzoor, R. "Targeted Adversarial Attack on Classic Vision Pipelines." *Computer Vision and Image Understanding*, Vol. 249, Art. 104140, Dec 2024.

Research Experience

Independent Researcher, LLM Agents and Test-Time Learning 2025 – Present
In collaboration with Stanford University

- Co-first author of *Epistemic Uncertainty for Test-Time Discovery* (arXiv:2605.11328, under review at ICLR 2027), joint work with Prof. John M. Cioffi's group at Stanford.
- Designed a mutual-information-shaped advantage with nuclear-norm regularization for test-time RL. It keeps about 2.2× more solution diversity than the clipped-IS baseline and improves maximum reward on 3 of 4 scientific-discovery benchmarks (Erdős minimum overlap, autocorrelation inequalities, GPU kernel optimization).
- Ran multi-node RL experiments on SLURM with models up to 120B parameters, across mathematics, GPU-kernel, algorithm-engineering and computational-biology benchmarks; designed and ran the ablation and robustness studies for peer review.
- Ongoing work (when-to-ask): testing whether divergence of mid-layer hidden states across trajectories can tell a long-horizon coding agent when to ask a human for input, earlier and more accurately than output-divergence baselines. Built the N-trajectory white-box harness, asynchronous hidden-state alignment, and the AUROC/lead-time evaluation suite.

Graduate Researcher / Research Assistant, Robotics & Machine Intelligence Lab 2020 – Jan 2023
NUST

- Led research on defending against adversarial attacks using state-of-the-art GANs, deep-learning denoisers and blind deconvolution.
- Designed targeted and deep black-box adversarial attacks on vision systems built with handcrafted features; analysed results statistically, qualitatively and quantitatively across robotics and computer-vision pipelines.
- Designed an automatic loop-closure detector evaluated on the KITTI dataset; prepared material for peer-reviewed journal submissions.

Work Experience

10Pearls Jun 2026 – Present

Staff Software Consultant (Agentic AI)

- Own the semantic/ontology layer and the agent runtime of AiX, an air-gapped operational platform in the Palantir Foundry class, delivered for Blackford Technologies (UAE). Wrote 20+ architecture decision records across both layers.
- Raised agent accuracy from 7/12 to 12/12, on three consecutive runs, on a 12-question benchmark over an OSDU subsurface dataset (95,260 objects, 243,957 edges). Added a deterministic multi-hop path planner and rewrote the tool descriptions the model sees, which fixed the two-hop reverse-traversal queries the agent could not answer before.
- Fixed the runtime reporting success on failed actions: the agent now checks each write by reading it back instead of retrying blindly, gets the real tool errors instead of sanitised ones, and can tell an invalid request from an upstream outage.

Wanclouds Inc. Sep 2023 – May 2026

AI Engineer

- Developed an autonomous AI agent for cloud-infrastructure monitoring, troubleshooting and automated issue resolution across multi-cloud environments, reducing downtime and manual work.
- Built a unit-testing framework for AI agents with DeepEval, run inside GitHub workflows; added a DSPy prompt optimizer to cut prompt-engineering time.
- Built a proprietary dataset and fine-tuned a small LLM on the company codebase as a privacy-compliant code assistant; brought training time down from 2.5 hours to 10–20 minutes.
- Ran adversarial evaluations with vulnerability scanners and patched security weaknesses in deployed LLMs; optimized fine-tuned LLMs to run on smaller GPUs and CPUs.

Macknowsoft Aug 2024 – Jan 2025

AI Consultant

- Built and delivered, as the sole developer, a production Arabic-language RAG chatbot for a Saudi client: Elasticsearch migration, embedding-based semantic search, answers with citations, and the full backend (APIs, authentication, PostgreSQL).

NetSol Technologies Jan 2023 – Sep 2023

Machine Learning Engineer

- Fine-tuned Stable Diffusion (DreamBooth, textual inversion) for consistent storybook characters; used ControlNet for a furniture-design mockup workflow.
- Trained YOLOv8 on a custom dataset for retail inventory tracking; built a hybrid CNN/rule-based classifier with 96% accuracy for radar-based patient fall detection.
- Led RFP development for a Defense Ministry Green-AI project proposing AI-based agricultural solutions.

Honours & Awards

- 3rd merit position, MS EE (AI & Autonomous Systems), NUST, 2022.
- Merit scholar (top undergraduate performer, tuition fee waiver), Air University, 2019.
- Certificate of Excellent Result, FSc Pre-Engineering, Army Public School and Colleges, 2016.

Leadership & Community

Co-lead, Build with AI track, Women Tech Quest 2026 2026

10Pearls' national competition for women in tech, about 1,000 participants (volunteer)

- Built the platform that scored submissions automatically on event day, set the track's evaluation criteria, mentored teams throughout and judged the finals.

Skills

ML / LLMs: PyTorch, TensorFlow, QLoRA/PEFT, bitsandbytes, DSPy, DeepEval, Hugging Face, OpenCV, scikit-learn, vector stores, SLURM multi-node training, LangGraph, agent tool contracts, agent evaluation harnesses

Engineering: Python, FastAPI, SQL (MySQL, PostgreSQL), Elasticsearch, Git, Linux, NumPy, Pandas, SciPy, Matplotlib, Seaborn, Pydantic, SQLAlchemy/Alembic, pgvector/Qdrant, Docker, Kubernetes

Certificates & Language: Generative AI with Large Language Models (Coursera); English, IELTS 7.5